

2019-2020 学年第一学期期末模拟练习

一、判断题(每小题 2 分, 共 20 分)

1. () For N elements, the *BuildHeap* function (which adjust an array of elements into a heap in linear time) does at most $N - \log(N+1)$ comparisons between elements.
2. () The best case time complexity of sorting algorithms based only on comparisons is at least $O(N \log N)$.
3. () Let P be the shortest path from S to T . If the weight of every edge in the graph is incremented by 2, P will still be the shortest path from S to T .
4. () Given that the pushing sequence of a stack is $\{1, 2, \dots, n\}$ and popping sequence is $\{x_1, x_2, \dots, x_n\}$. If $x_2 = n$, we can obtain $n-1$ different possible popping sequences.
5. () To judge an odd integer $N (> 10)$ is prime or not, we need to check if it is divisible by any odd number from 3 to $\sqrt{N}$. The time complexity of this algorithm is $O(0.5 \log N)$.
6. () In a binary tree, if node A is an ancestor of node B , then A must precede B in the inorder traversal sequence.
7. () During finding all articulation points in a connected graph, if v_1 is the parent of v_2 in the depth-first tree, $\text{Num}(v_1)$ must be less than $\text{Num}(v_2)$.
8. () During the sorting, processing every element which is not yet at its final position is called a "run". To sort a list of integers using quick sort, it may reduce the total number of recursions by processing the small partition first in each run.
9. () The number of leaf nodes in a binary tree is only related to the number of 2-degree nodes, i.e. it has nothing to do with the number of 1-degree nodes.
10. () There must be a collision if we insert a new element into a hash table with the loading density being

二、选择题(每小题 3 分, 共 60 分)

1、The recurrent equations for the time complexities of programs P1 and P2 are:

- P1: $T(1)=1, T(N)=T(N/3)+1$
- P2: $T(1)=1, T(N)=3T(N/3)+1$

Then the correct conclusion about their time complexities is: ()

Q. they are both $O(\log N)$ A, B, C, D, I

R. $O(\log N)$ for P1, $O(N)$ for P2 A, B, I

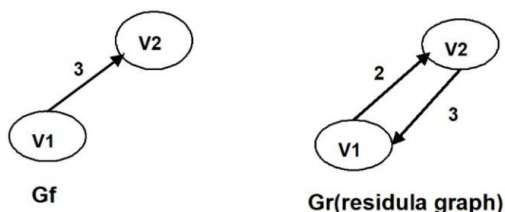
S. they are both $O(N)$

T. $O(\log N)$ for P1, $O(N \log N)$ for P2

2、 Given input { 321, 156, 57, 46, 28, 7, 331, 33, 34, 63 }. Which one of the following is the result after the 2nd run of the Least Signification Digit (LSD) radix sort? ()

- I. $\rightarrow 7 \rightarrow 321 \rightarrow 28 \rightarrow 34 \rightarrow 33 \rightarrow 331 \rightarrow 156 \rightarrow 46 \rightarrow 57 \rightarrow 63$
- J. $\rightarrow 7 \rightarrow 321 \rightarrow 28 \rightarrow 331 \rightarrow 34 \rightarrow 33 \rightarrow 46 \rightarrow 156 \rightarrow 57 \rightarrow 63$
- K. $\rightarrow 156 \rightarrow 28 \rightarrow 321 \rightarrow 331 \rightarrow 33 \rightarrow 34 \rightarrow 46 \rightarrow 57 \rightarrow 63 \rightarrow 7$
- L. $\rightarrow 7 \rightarrow 321 \rightarrow 28 \rightarrow 331 \rightarrow 33 \rightarrow 34 \rightarrow 46 \rightarrow 156 \rightarrow 57 \rightarrow 63$

3、 When solving the maximum flow problem for graph G, if partial states of the Gf (will be the maximum flow when the algorithm terminates) and Gr (residual graph) are shown as the following, what must be the capacity of (v1, v2) or of (v2,v1) in the original graph G? ()



- Q. the capacity of (v1, v2) is 2
- R. the capacity of (v1, v2) is 3
- S. the capacity of (v1, v2) is 5
- T. the capacity of (v2, v1) is 5

4、 Given a tree of degree 5. Suppose that the numbers of nodes of degrees 1, 2, 3, 4, 5 are 3, 4, 2, 1, 3, respectively. Then the number of leaf nodes must be _____. ()

- Q. 30
- R. 29
- S. 23
- T. 24

5、 A graph with 90 vertices and 20 edges must have at least __ connected component(s). ()

- Q. 69
- R. 70
- S. 84
- T. 85

6、 The time complexity of traversing a binary tree with N nodes and height H in levelorder (with the help of with an auxiliary queue) is _____. ()

- I. $O(N)$
- J. $O(H)$

K. $O(M\log H)$

L. $O(H+N)$

7、The array representation of the disjoint sets is given by $\{ 3, 1, -5, 2, 1, -3, -1, 6, 6 \}$. Keep in mind that the elements are numbered from 1 to 9. After invoking $\text{Union}(\text{Find}(4), \text{Find}(8))$ with union-by-size, which elements will be changed in the resulting array? ()

Q. elements of index 3 and 6

R. elements of index 4 and 8

S. elements of index 6 and 1

T. elements of index 4 and 7

8、Given an integer sequence 25、84、21、47、15、27、68、35、20, after the first run of the shell sorting by an increment 3, the integer sequence is _____. ()

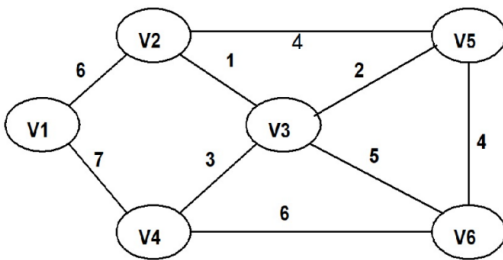
A. 25 15 20 27 35 21 68 84 47

B. 15 21 20 25 84 27 68 35 47

C. 25 15 20 47 21 35 68 84 27

D. 25 15 20 47 35 21 68 84 27

9、To find the minimum spanning tree with Kruskal's algorithm for the following graph. Which edge will be added in the final step? ()



Q. $(v1, v4)$

R. $(v1, v2)$

S. $(v4, v6)$

T. uncertain

10、Suppose that the range of a hash table is $[0, 16]$, and the hash function is $H(\text{Key}) = \text{Key} \% 13$. If linear probing is used to resolve collisions, then after inserting $\{ 24, 12, 13, 49, 23 \}$ one by one into the hash table, the index of 23 is: ()

I. 13

J. 0

K. 10

L. 1

11、 Given a binary search tree with its preorder traversal sequence { 10, 8, 21, 12, 15, 32 }. If 10 is deleted from the tree, which one of the following statements is FALSE? ()

- I. One possible preorder traversal sequence of the resulting tree may be { 8, 21, 12, 15, 32 }
- J. One possible preorder traversal sequence of the resulting tree may be { 21, 12, 8, 15, 32 }
- K. One possible preorder traversal sequence of the resulting tree may be { 12, 8, 21, 15, 32 }
- L. It is possible that the new root may have 2 children

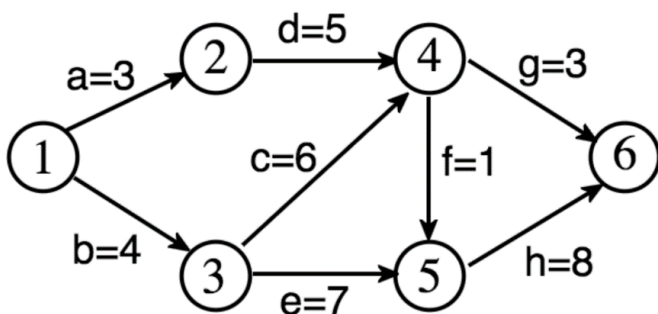
12、 For an in-order threaded binary tree, if the post-order and in-order traversal sequences are DFGEBCA and DBFEGAC respectively, the number of left threaded links and that of right threaded links are _____. ()

- Q. 5 and 3
- R. 4 and 4
- S. 4 and 3
- T. 3 and 5

13、 To list the directory in a hierarchical file system with the format of files that are of depth d_i will have their names indented by d_i tabs, which of the following traversals is the most suitable one. ()

- E. pre-order
- F. in-order
- G. post-order
- H. level-order

14、 Given the AOE network of a project with 8 activities shown below, the earliest/latest completion time for node 4 is _____. ()



- E. 8 and 8
- F. 10 and 16
- G. 10 and 10
- H. 8 and 14

15、 Suppose that the level-order traversal sequence of a max-heap is $\{78, 72, 53, 42, 64, 36\}$. Use the linear algorithm to adjust this max-heap into a min-heap, and then insert 38. The postorder traversal sequence of the resulting tree is: ()

E. 72, 42, 64, 36, 78, 38, 53

F. 72, 64, 42, 78, 53, 38, 36

G. 78, 64, 53, 72, 42, 38, 36

H. 53, 64, 38, 72, 78, 42, 36

16、 Insert $\{20, 25, 13, 22, 4, 9, 29, 35, 14, 17\}$ one by one into an initially empty hash table of size 13 with the hash function $H(\text{Key}) = \text{Key} \% 13$, and quadratic probing is used to resolve collisions. How many numbers can be inserted without collisions? ()

E. 5

F. 6

G. 7

H. 8

17、 For a complete binary tree with odd number of nodes, among the following statements, how many statement(s) is/are true? ()

a) the height of the left subtree must be greater than that of the right subtree

b) the number of nodes in the left subtree may be greater than that in the right subtree

c) the number of leaf-nodes in the left subtree must be twice over that in the right subtree

d) the number of leaf-nodes and that of non-leaf-nodes may be the same

E. 1

F. 2

G. 3

H. 4

18、 Graph G is an undirected completed graph of 20 nodes. Is there an Euler circuit in G? If not, in order to have an Euler circuit, what is the minimum number of edges which should be removed from G? ()

E. Yes, Graph G has an Euler circuit

F. No, Graph G has no Euler circuit. 10 edges should be removed.

G. No, Graph G has no Euler circuit. 20 edges should be removed.

H. No, Graph G has no Euler circuit. 40 edges should be removed.

19、 If keys are pushed onto a stack in the order **abcdef**, then it's impossible to obtain the output sequence ()

E. fedcba

F. bcafed

G. dcefbfa

H. cabdef

20、The space complexity of traversing a binary tree with N nodes and height H in levelorder (with the help of an auxiliary queue) is _____. ()

E. $O(H+N)$ F. $O(N)$ G. $O(H)$ H. $O(N+\log H)$

三、程序填空题（满分 12 分）

1、(6 分) The function Unweighted is to find the unweighted shortest path from Vertex S to every other vertices in a given Graph. The distances are stored in dist[], and path[] records the paths. The Graph is defined as the following:

```
typedef struct GNode *PtrToGNode;
struct GNode{
    int Nv;           /* Number of vertices */
    int Ne;           /* Number of edges */
    AdjList List;     /* adjacency matrix */
};
typedef PtrToGNode Graph;

void Unweighted( Graph G, Queue Q, int dist[], int path[], Vertex S )
{
    Vertex V, U;
    NodePtr ptr;

    dist[S] = 0;
    Enqueue(S, Q);
    while ( !IsEmpty(Q) ) {
        V = Dequeue( Q );
        for ( ptr=G->List[V].FirstEdge; ptr; ptr=ptr->Next) {
            U = ptr->AdjV;
            if ( dist[U] == INFINITY ) {
                _____;(3 分)

                path[U] = V;
                _____;(3 分)
            }
        }
    }
}
```

2、（6分）None-recursive Quick Selection algorithm

The function is to find the K-th smallest element in a list A of N elements. The initial function call is QSelect(A, N, K). Please complete the following program.

ElementType QSelect(ElementType A[], int N, int K)

```
{
    ElementType Pivot;
    int L, R, Left, Right, K1;
    Left = 0; Right = N-1; K1 = K;
    for(;;) {
        L = Left, R = Right+1;
        Pivot = A[Left];
        while (1) {
            while ( A[++L] < Pivot ) ;(3 分)
            if ( L < R ) Swap( &A[L], &A[R] );
            else break;
        }
        Swap( &A[Left], &A[R] );
        if ( K1 < (L-Left) )
            Right = R-1;
        else if ( K1 > (L-Left) ){
            _____;(3 分)
            Left = L;
        }else
            return Pivot;
    }
}
```

四、函数题（8分）

Is It CBST

Insert a sequence of integer keys into an initially empty binary search tree with **larger** keys in its **left** subtree and **smaller** keys in its **right** subtree, you are supposed to tell if the resulting tree is a **complete** binary search tree (CBST), and then return its postorder traversal sequence.

Format of function:

int IsCBST(int A[], int N);

where the sequence of N insertions is stored in A[].

The function IsCBST is supposed to return 1 if the resulting tree is a CBST, or 0 if not. At the mean time, A[] must store the postorder traversal sequence of the resulting tree.

Sample program of judge:

```
#include <stdio.h>
#define MAXN 10
int IsCBST( int A[], int N );
int main()
{
    int A[MAXN], N, i;
    scanf("%d", &N);
    for (i=0; i<N; i++) scanf("%d", &A[i]);
    if ( IsCBST(A, N) ) printf("Yes\n");
```

```
    else printf("No\n");  
    for (i=0; i<N; i++) printf("%d ", A[i]);  
    return 0;  
}  
/* Your function will be put here */
```

Sample Input 1:

9
38 45 42 24 58 30 67 12 51

Sample Output 1:

Yes
67 51 58 42 45 30 12 24 38

Sample Input 2:

8
38 24 12 45 58 67 42 51

Sample Output 2:

No
67 51 58 42 45 12 24 38

2019-2020 学年第一学期期末模拟练习参考答案

一、判断题(每小题 2 分, 共 20 分)

1、【正解】F

【解析】在堆排序中先进行一次堆的线性初始化操作, 其时间复杂度为 $O(n)$, 最多需要比较的次数为 $2*(n-\log n)$

【考点延伸】堆排序

2、【正解】F

【解析】基于比较的排序算法有冒泡排序, 简单选择排序等, 时间复杂度均为 $O(N^2)$

【考点延伸】排序时间复杂度

3、【正解】F

【解析】从 S 到 T 的最短路径 P 可能包含很多条边, 虽然权值比较小, 但当 每条边的权值都增加 2 时, 此时 P 就不一定是最短路径了

【考点延伸】最短路径

4、【正解】T

【解析】第二个出栈的元素是入栈序列中的最后一个, 这说明前 $n-1$ 个元素中有一个最先出栈, 然后 n 出栈, 所以后面的出栈序列是确定的, 因此能够得到 $n-1$ 个出栈序列。

【考点延伸】栈

5、【正解】F

【解析】最坏需要遍历 $(\sqrt{N}-3)/2$ 次, 因此时间复杂度是 $O(\sqrt{N})$

【考点延伸】时间复杂度

6、【正解】F

【解析】A 是 B 的祖先, 但 B 可能在 A 的左子树或右子树上, 因此在中序遍历二叉树时, A 不一定在 B 的前面

【考点延伸】二叉树的遍历

7、【正解】T

【解析】二叉树采用深度优先搜索建树, 则如果 V_1 是 V_2 的父结点, 则 V_1 被遍历的次数一定少于 V_2

【考点延伸】深度优先搜索

8、【正解】F

【解析】快速排序的思想是找到一个枢纽, 然后使其左边的元素小于它, 右边的元素大于它, 再分别递归处理左右子区间。每次运行时先处理小的部分不会减少递归次数

【考点延伸】快速排序

9、【正解】T

【解析】邻接表是结构体数组加链表的结合体, 数组存储顶点, 每一个顶点的链表存储与该顶点

相连的所有顶点，边越多，需要开辟的链表越长，所需存储空间越大。

【考点延伸】邻接表

10、【正解】T

【解析】根据二叉树的特点，边数+1=节点数，度为1的结点数+度为2结点数*2=边数，利用两个公式可得叶子结点数仅和度为2的结点有关

【考点延伸】二叉树

二、单选题(每小题3分，共60分)

1 【正解】B

【解析】通过迭代可以计算出 $P1: T(n) = \log_3 n + 1, P2: T(n) = \frac{3}{2}n - \frac{1}{2}$ 因此 $O(\log N)$ for

$P1, O(N)$ for $P2$

【考点延伸】时间复杂度

2 【正解】D

【解析】经过一次次位优先排序结果为：321 331 33 63 34 156 46 57 7 28

第二次结果为：7 321 28 331 33 34 46 156 57 63

【考点延伸】次位优先排序

3 【正解】C

【解析】在 G_f 中添加路径，在 G_r 中反向添加，因此添加了一个 1-2 的路径，反向添加了 2-3 的路径，但是最小流量是 3， G_r 中的总流量是不变的，是 5

【考点延伸】网络流

4 【正解】D

【解析】 $\text{leafnodes} + 3 + 4 + 2 + 1 + 3 = \text{边数} + 1$ ， $1 * 3 + 2 * 4 + 3 * 2 + 4 * 1 + 5 * 3 = \text{边数}$ ，故 $\text{leafnodes} = 24$

【考点延伸】树

5 【正解】B

【解析】要使连通体最小，则 20 条边应连接最多的顶点，即 21，最小生成树，所以至少有 $90 - 21 + 1 = 70$ 个

【考点延伸】连通体

6 【正解】A

【解析】从访问的结点来看，共需访问 n 个结点，因此时间复杂度为 $O(N)$

【考点延伸】层序遍历

7 【正解】A

【解析】4 所在集合的结点为 3，size 为 5，8 所在集合的结点为 6，size 为 3，6 会连接在 3 上，因此 6 的父结点会变化，3 的 size 会变化

【考点延伸】并查集

8 【正解】D

【解析】增量为 3，每隔 3 个的序列排序，第一组是 25 47 68，排序后为 25 47 68 第二组是 84 15 35，排序后为 15 35 84，第三组是 21 27 20，排序后为 20 21 27，合并一下最后结果就是 25 15 20 47 35 21 68 84 27

【考点延伸】希尔排序

9 【正解】B

【解析】根据 Kruskal's algorithm，依次选择的顺序为 2-3，3-5，3-4，5-6，1-2

【考点延伸】最短路径

10 【正解】A

【解析】13 在 0，49 在 10，24 在 11，12 在 12，23 在 13

【考点延伸】散列表

11 【正解】B

【解析】中序遍历为：8 10 12 15 21 32，根结点被删除后，8 或者 12 可以替换根结点，因此 A,C,D 正确

【考点延伸】二叉搜索树

12 【正解】B

【解析】由前序和后序遍历可得二叉树的结构，共有 4 个叶子结点，因此左链接数和右链接数都为 4

【考点延伸】线索树

13 【正解】A

【解析】采用前序遍历会先输出根结点，相当于缩进了 d_i 个 tabs

【考点延伸】二叉树的遍历

14 【正解】C

【解析】经过两次拓扑排序可得 1 的最早开始时间为 0，最晚开始时间为 0；2 的最早开始时间为 3，最晚开始时间为 5；3 的最早开始时间为 4，最晚开始时间为 4；4 的最早开始时间为 10，最晚开始时间为 10；5 的最早开始时间为 11，最晚开始时间为 11；6 的最早开始时间为 19，最晚开始时间为 19；

【考点延伸】拓扑排序

15 【正解】B

【解析】采用筛选法来调整成最小堆，时间复杂度为 $O(n)$ ，调整后的层序遍历结果为 36 42 53 72 64 78，插入 38 后的层序结果为 36 42 38 72 64 78 53，故后序遍历为 72 64 42 78 53 38 36

【考点延伸】堆

16 【正解】C

【解析】不会发生冲突的有 13 14 29 4 20 22 25

【考点延伸】散列冲突

17 【正解】A

【解析】当结点为 6 时，左子树和右子树树高相等，并且叶子结点数和非叶子结点数相等；左子树叶子结点数并不一定是右子树结点数的两倍多

【考点延伸】完全二叉树

18 【正解】B

【解析】如果一个连通图的所有顶点的度数都为偶数，那么这个连通图具有欧拉回路，且这个图被称为欧拉图。具有 20 个结点的完全无向图，每个顶点的度数都为 19。要成为欧拉图，则需要让各顶点的度数至少减 1，因此最少需要去除 10 条边，一条边连接 2 个顶点，度变为 18。

【考点延伸】欧拉图

19 【正解】D

【解析】可以将出栈序列逆序放入栈中，在另一个栈中依次按入栈序列进栈，如果栈顶相同就出栈，最后检查两个栈是否同为空，如果都为空，则符合要求。

【考点延伸】栈

20 【正解】B

【解析】 n 个元素都有机会入队和出队，因此额外消耗的空间复杂度为 $O(n)$

【考点延伸】空间复杂度

三、程序填空题（满分 12 分）

1、【解析】(1) $\text{dist}[U]=\text{dist}[V]+1$ //从 V 到 U 存在最短路径,因此修改 U 到 S 的距离 (3 分)

(2) $\text{Enqueue}(U,Q)$ //将 U 收录进已确定最短顶点集合 (3 分)

【考点延伸】无权最短路径

2、【解析】(1) $\text{while}(A[--R]>\text{pivot})$ //找到 pivot 右边小于等于 pivot 的元素下标 (3 分)

(2) $K1=K1-L+\text{Left}$ //此时 K1 在 L 的右边，需要找第 $K1-L+\text{Left}$ 小的元素(3 分)

【考点延伸】快速排序

四、函数题(满分 8 分)

【解析】

```
/* Your function will be put here */
```

void Insert(int *ret, int num,int index){//数值插入函数但是由于题目数组为 0-n-1 的，所以要思考左右下标

```
int left = (index+1)*2-1,right = (index+1)*2;
```

```
if(!ret[index])ret[index] = num;
```

```
else {
```

```
    if(num>ret[index]) Insert(ret,num,left);
```

```
    else Insert(ret,num,right);
```

```
}
```

```
return;
```

```
}
```

```
int cnt=0;
```

```
void PostOrder(int *ret, int *a, int index){
```

```
    if(ret[index]){
```

```
        PostOrder(ret,a,(index+1)*2-1);
```

```
        PostOrder(ret,a,(index+1)*2);
```

```
        a[cnt++] = ret[index];
```

```
    }
```

```
}
```

```
int IsCBST( int A[], int N ){
```

```
    int ret[111]={0};
```

```
    for(int i=0; i<N; i++) Insert(ret,A[i],0);
```

```
    PostOrder(ret,A,0);
```

```
    int flag=1;
```

for(int i =0; i<N; i++)if(!ret[i])flag = 0;//如果是完全二叉树，则所有的点都在要求的 N 中，而不是则不然：在 ret[x](x>N)的条件下会有节点存在

```
    return flag;
```

```
}
```

【考点延伸】二叉树